



Design of Local Area Network

Communication Network Engineering

EN2150

Department of Electronic and Telecommunication Engineering
University of Moratuwa

230321P	A.H.D. Karunanayake
230458P	H.H. Palihena
230493R	N.P.P. Piyumal
230613M	M.N.N. Shehan

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1 Introduction

This report presents the design of the backbone network for the University of Moratuwa and the internal network of the ENTC building. The objective is to develop a scalable, reliable, and cost-effective network infrastructure suitable for long-term use (20–25 years).

2 Backbone Network Design

2.1 Design Approach and Topology

The backbone network of the University of Moratuwa is designed using a hierarchical architecture consisting of core and distribution layers. This approach is selected to ensure scalability, manageability, efficient traffic handling and error isolating across the campus network.

The Center for Information Technology Services (CITeS) acts as the central core of the network. Two multilayer core switches are deployed at CITeS to provide high availability and eliminate single point failures. These core switches are interconnected and operate in a redundant configuration. The Cisco 3650-24PS multilayer switch is selected here due to its advanced Layer 3 capabilities, high performance and scalability. As the core of the network is responsible for handling aggregated traffic from all buildings, it requires a device capable of both high speed switching and efficient routing.

A set of distribution switches is used to aggregate connections from multiple buildings. Each distribution switch is connected to both core switches, ensuring continuous connectivity even in the event of a link or device failure. This redundancy improves the reliability and robustness of the network. The connection between the core and distribution layers is implemented using high-speed Single Mode Fiber (SMF) links supporting bandwidths of up to 10 Gbps. This choice is made because the core layer handles aggregated traffic from multiple buildings, resulting in a very high data load. Fiber optic cables are preferred over copper in this segment due to their higher bandwidth capacity, low signal attenuation and ability to support long distance communication across the campus.

All academic and administrative departments are connected to a distribution switch. This structured approach avoids direct interconnections between buildings, reducing network complexity and cabling distance. Distribution switch and department switches are connected using Single Mode Fiber or Gigabit Ethernet over copper depending on the traffic demand of each department. For high demand departments such as ENTC, CSE, Electrical fiber optic links (Single Mode Fiber) are used to provide higher bandwidth and support large volumes of data generated by laboratories, research activities and high user density. Fiber is selected in these cases due to its high capacity, low latency and ability to support future bandwidth upgrades. For departments with relatively lower traffic requirements such as library, Fashion Design Gigabit Ethernet over copper is used. Copper cabling provides sufficient bandwidth (1 Gbps) for typical academic and administrative activities while being more cost effective and easier to deploy compared to fiber.

Overall, the selected topology ensures a balance between performance, cost and scalability, making it suitable for long-term deployment over the next 20–25 years.

2.2 Network Topology Diagram

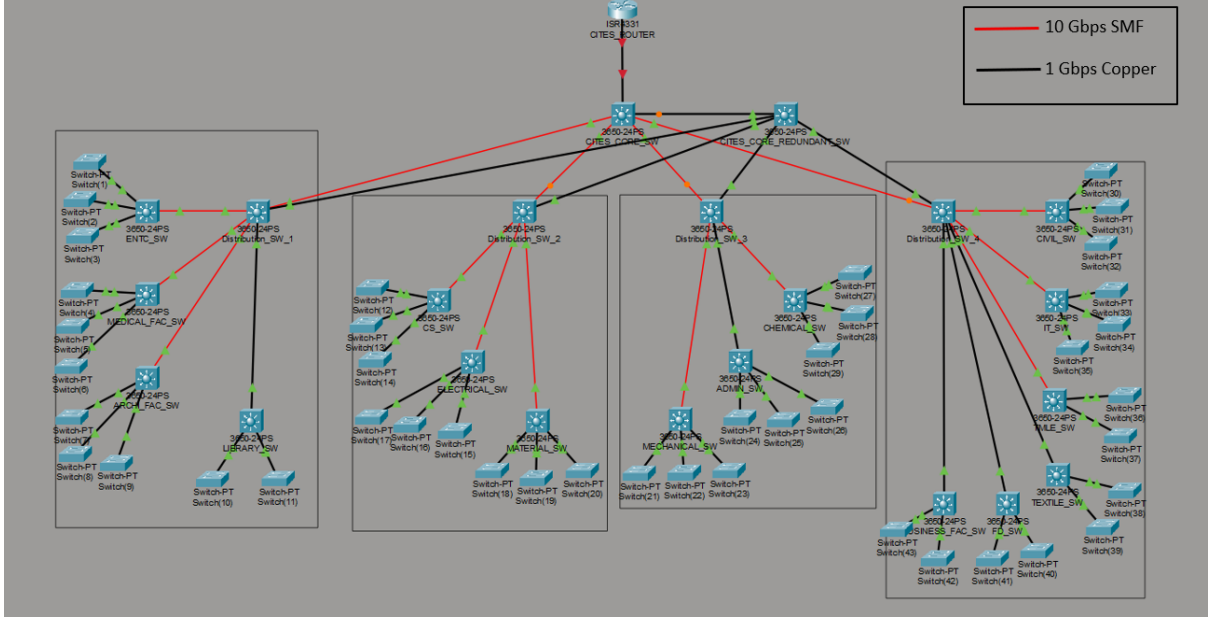


Figure 1: Backbone Network Topology

2.3 Bandwidth Allocation

The bandwidth allocation in the backbone network is designed based on the expected traffic load at different layers of the network. Since traffic increases as it moves towards the core due to aggregation from multiple buildings, higher bandwidth links are allocated at upper layers.

The links between the core and distribution layers are provisioned with high-capacity connections (10 Gbps using Single Mode Fiber) to handle aggregated traffic from all connected buildings. Between the distribution layer and department switches, bandwidth is allocated based on the demand of each department.

3 IP Addressing Scheme

3.1 IPv4 Addressing

3.1.1 Department Addressing

Department	IPv4 Network	Subnet Mask
ENTC	192.168.1.0	/24
Medical Faculty	192.168.2.0	/24
Archi Faculty	192.168.3.0	/24
Library	192.168.4.0	/24
CSE	192.168.5.0	/24
Electrical	192.168.6.0	/24
Material	192.168.7.0	/24
Mechanical	192.168.8.0	/24
Chemical	192.168.9.0	/24
Administration Building	192.168.10.0	/24
Civil	192.168.11.0	/24
IT Faculty	192.168.12.0	/24
TMLE	192.168.13.0	/24
Textile	192.168.14.0	/24
Fashion Design	192.168.15.0	/24
Business Faculty	192.168.16.0	/24

3.1.2 Backbone Link Addressing

Link	Network
Core1 - Core2	10.0.0.0/30
Core1 - Dist1	10.0.0.4/30
Core2 - Dist1	10.0.0.8/30
Core1 - Dist2	10.0.0.12/30
Core2 - Dist2	10.0.0.16/30
Core1 - Dist3	10.0.0.20/30
Core2 - Dist3	10.0.0.24/30
Core1 - Dist4	10.0.0.28/30
Core2 - Dist4	10.0.0.32/30

3.2 IPv6 Addressing

Building	IPv6 Network
ENTC	2001:db8:1::/64
Medical Faculty	2001:db8:2::/64
Archi Faculty	2001:db8:3::/64
Library	2001:db8:4::/64
CSE	2001:db8:5::/64
Electrical	2001:db8:6::/64
Material	2001:db8:7::/64
Mechanical	2001:db8:8::/64
Chemical	2001:db8:9::/64
Administration Building	2001:db8:10::/64
Civil	2001:db8:11::/64

IT Faculty	2001:db8:12::/64
TMLE	2001:db8:13::/64
Textile	2001:db8:14::/64
Fashion Design	2001:db8:15::/64
Business Faculty	2001:db8:16::/64

3.3 Justification

This IP addressing scheme is designed to ensure efficient network organization, scalability and ease of management. A hierarchical addressing approach is used where each building in the university is assigned a unique subnet within the private IPv4 address range.

The use of the 192.168.x.0/24 addressing format provides sufficient host capacity for each building while allowing straightforward identification of network segments based on the subnet value.

For backbone connections between core and distribution devices, smaller subnets are used to optimize IP address utilization. This prevents wastage of address space while still supporting point-to-point communication.

In addition to IPv4, IPv6 addressing is incorporated using a structured prefix scheme. Each building is assigned a unique /64 subnet, which is the standard for IPv6 networks. This ensures compatibility with modern networking requirements and supports future scalability.

4 ENTC Internal Network Design

4.1 LAN Structure

In our LAN design, the network can be divided into two main categories. The first is how lab PCs connect to the internet, and the second is how students' laptops and mobile devices connect wirelessly.

In the first case, a student PC connects to a 24-port access layer switch using a copper UTP cable through a patch panel and wall socket. The patch panel is used for structured cabling and easier management. The access switch then connects to the ENTC distribution/core switch using a fiber optic link (typically through Gigabit Ethernet uplink ports). This core switch is connected to the ENTC backbone network, which provides access to other networks and the internet.

In the second case, students' laptops and mobile phones connect wirelessly to a wireless access point (AP-PT). The access point is connected to a 24-port access switch using a copper Ethernet cable. Similar to the first case, this access switch is connected to the ENTC distribution/core switch via a fiber optic link. Through this path, wireless devices gain access to the backbone network and the internet.

4.2 Network Diagram

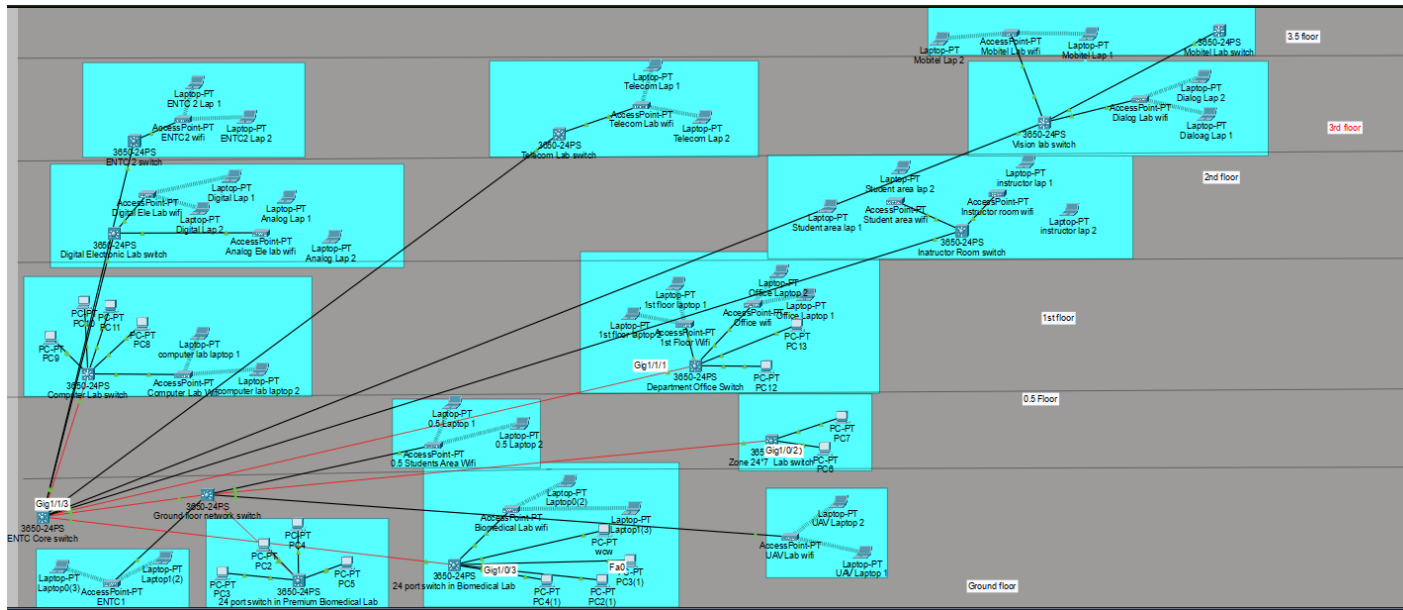


Figure 2: ENTC LAN

5 Active and Passive Components

For component selection, we used Cisco Catalyst 3650-24PS switches because they support fiber uplink connections (via SFP ports) and provide high performance suitable for distribution/core operations. These switches also support advanced features such as VLANs and Layer 3 routing, which are useful for efficient network management.

For cabling, we used single-mode fiber optic cables for high-speed, long-distance communication between switches (especially for backbone and uplink connections). For short-distance communication, such as connecting PCs and access points to switches, we used copper UTP cables (e.g., Cat5e/Cat6) due to their lower cost and ease of installation.

We also used network racks to house the access layer switches and the core/distribution switches. This ensures proper physical protection, organized cable management, better airflow for cooling, and easier maintenance of network components.

6 Bill of Quantities (BOQ)

6.1 Active Components

Item	Quantity	Description
Core Switch (3650)	2	Core Layer

Distribution Switch	4	Distribution Layer
Access Switch	16	Buildings

Table 4: Backbone Network Design

Item	Quantity	Description
Switches	13	3650-24PS for core switch and distribution switches
Wireless access points	15	AP-PT devices

Table 5: ENTC LAN Design

6.2 Passive Components

Item	Quantity	Description
Fiber Cable	5 km approximately	Core switch to distribution switches
Copper UTP Cable	5 km approximately	distribution switches to PCs
Patch Panels	40	To organize cables in network racks
Network racks	40	To store switches, patch panels, internal cables

Table 6: Passive Component BOQ

7 Simulation Results

7.1 For the backbone network

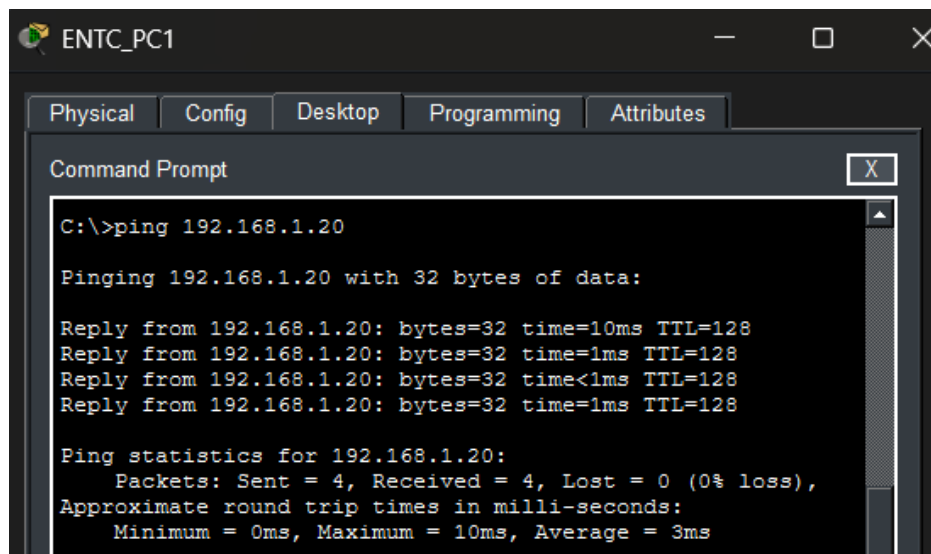
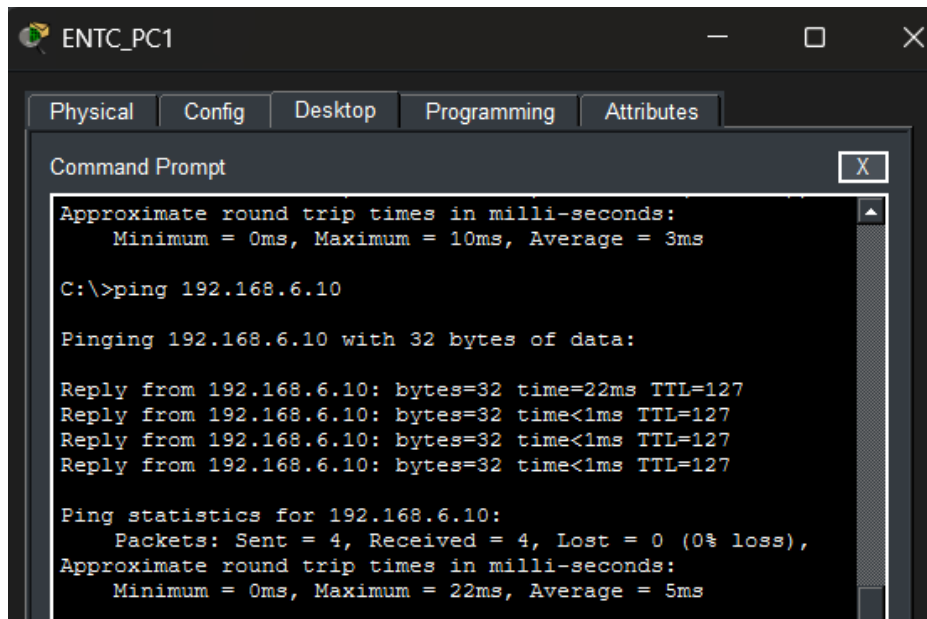


Figure 3: Same VLAN Communication (ENTC PC1 with ENTC PC2)



The screenshot shows a Windows Command Prompt window titled "ENTC_PC1". The window has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Config" tab is active. The Command Prompt displays the following text:

```
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms

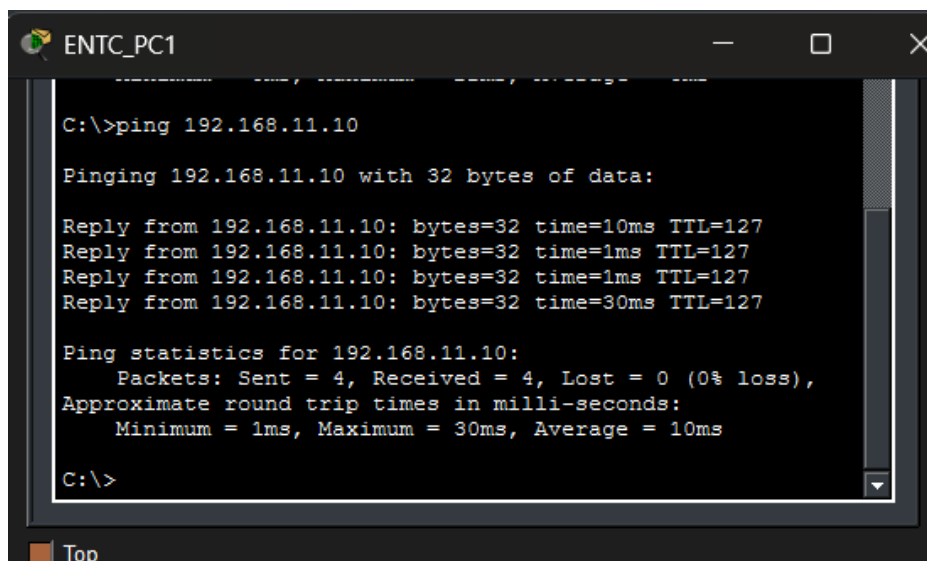
C:\>ping 192.168.6.10

Pinging 192.168.6.10 with 32 bytes of data:

Reply from 192.168.6.10: bytes=32 time=22ms TTL=127
Reply from 192.168.6.10: bytes=32 time<1ms TTL=127
Reply from 192.168.6.10: bytes=32 time<1ms TTL=127
Reply from 192.168.6.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.6.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 22ms, Average = 5ms
```

Figure 4: Inter-VLAN Communication (ENTC PC1 with Electrical PC1)



The screenshot shows a Windows Command Prompt window titled "ENTC_PC1". The window has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Config" tab is active. The Command Prompt displays the following text:

```
C:\>ping 192.168.11.10

Pinging 192.168.11.10 with 32 bytes of data:

Reply from 192.168.11.10: bytes=32 time=10ms TTL=127
Reply from 192.168.11.10: bytes=32 time=1ms TTL=127
Reply from 192.168.11.10: bytes=32 time=1ms TTL=127
Reply from 192.168.11.10: bytes=32 time=30ms TTL=127

Ping statistics for 192.168.11.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 30ms, Average = 10ms

C:\>
```

Figure 5: Inter-VLAN Communication (ENTC PC1 with Civil PC1)

7.2 For the ENTC LAN network

```
Cisco Packet Tracer PC Command Line 1.0
C:\>

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 6: Verifying the working of the computer lab switch by checking the connection between two PCs in the Computer lab

Switch>show interfaces status						
Port	Name	Status	Vlan	Duplex	Speed	Type
Gig1/0/1		connected	1	a-full	a-100	10/100/1000BaseTX
Gig1/0/2		connected	1	a-full	a-100	10/100/1000BaseTX
Gig1/0/3		connected	1	a-full	a-100	10/100/1000BaseTX
Gig1/0/4		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/5		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/6		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/7		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/8		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/9		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/10		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/11		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/12		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/13		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/14		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/15		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/16		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/17		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/18		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/19		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/20		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/21		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/22		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/23		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/24		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/1/1		connected	1	a-full	a-100	10/100/1000BaseTX
Gig1/1/2		connected	1	a-full	a-100	10/100/1000BaseTX
Gig1/1/3		connected	1	a-full	a-100	10/100/1000BaseTX
Gig1/1/4		notconnect	1	auto	auto	10/100/1000BaseTX

Figure 7: Core switch interfaces

Switch>show interfaces status						
Port	Name	Status	Vlan	Duplex	Speed	Type
Gig1/0/1		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/2		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/3		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/4		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/5		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/6		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/7		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/8		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/9		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/10		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/11		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/12		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/13		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/14		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/15		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/16		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/17		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/18		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/19		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/20		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/21		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/22		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/23		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/0/24		notconnect	1	auto	auto	10/100/1000BaseTX
Gig1/1/1		connected	1	a-full	a-100	10/100/1000BaseTX
Gig1/1/2		connected	1	a-full	a-100	10/100/1000BaseTX
Gig1/1/3		connected	1	a-full	a-100	10/100/1000BaseTX
Gig1/1/4		connected	1	a-full	a-100	10/100/1000BaseTX

Figure 8: Ground floor switch interfaces

Figure 9: Verifying core switch to ground floor distribution switch connection

7.3 Link to Packet Tracer Simulation files

All simulation files can be found in this link.

<https://drive.google.com/drive/folders/19PSJvjeE3SOBPM-LwPL0D5eEzM7dn981?usp=sharing>

8 Conclusion

The proposed network design provides a scalable, reliable, and cost-effective solution for the University of Moratuwa as well as for the ENTC LAN. The use of hierarchical

architecture and redundancy ensures long-term sustainability and performance.